



DR ESLAM AHMED
eliteigcse.com

Algebraic Fractions

Factorise everything first. Then it is just fractions: cancel, flip, or find the common denominator.

THE TOPIC ON ONE MAP

Algebraic Fractions

01 Simplify: cancel brackets

$$\frac{x^2 - 9}{x^2 + x - 6} \quad (3 - x) \text{ vs } (x - 3)$$

02 Multiply and divide: KFC

× cancel first ÷ Keep-Flip-Change

03 Add and subtract: the LCM

"as a single fraction"

04 Mixed operations

BODMAS fractions inside fractions

05 Solving with fractions

× every term by the LCM reject negative lengths

THE GOLDEN RULE

The first step of **any** algebraic fraction question:
factorise every top and every bottom.

01 Simplifying: Cancel Brackets

After factorising, cancel a bracket that is on the top **and** the bottom. Whole brackets only.

WHEN YOU SEE → YOU DO

“simplify fully” (fraction)

→ golden rule: factorise everything

$(3 - x)$ against $(x - 3)$

→ take out -1 , then cancel

WORKED MODEL • FACTORISE, THEN CANCEL

$$\frac{x^2 - 16}{2x^2 - 5x - 12}$$

$$\textcircled{1} \quad \frac{(x + 4)(x - 4)}{(2x + 3)(x - 4)}$$

DOTS on top • CC Rule below

$$\textcircled{2} \quad \frac{(x + 4)(\cancel{x - 4})}{(2x + 3)(\cancel{x - 4})}$$

cancel the same bracket

ANSWER $\frac{x + 4}{2x + 3}$

× **TRAP** You cannot cancel *terms*: $\frac{x + 4}{x + 6}$ is finished. Only a bracket that **multiplies** everything can go.

OPPOSITE BRACKETS $(3 - x) = -(x - 3)$. Take out the minus, then cancel.

IDEA 01

Practise: Simplifying: Cancel Brackets

My questions first, then selected practice.

Q1 DR ESLAM Golden rule first

CH 2 · TOPIC 6 3 marks

Simplify fully $\frac{x^2 - 9}{x^2 + x - 6}$.

Top: DOTS. Bottom: CC Rule. Then cancel.

Answer _____

Q2 DR ESLAM HCF on top

CH 2 · TOPIC 6 2 marks

Simplify fully $\frac{x^2 + 5x}{x^2 - 25}$.

Answer _____

SELECTED PRACTICE

Q3 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 2 marks

4 Simplify $\frac{4(x+5)}{x^2 + 2x - 15}$

(2 marks)

Q4 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 1 mark

14 Circle the expression that is equivalent to $\frac{3x^2}{6x^2 + 3}$

$\frac{x^2}{2x^2 + 3}$	$\frac{x^2}{6x^2 + 1}$	$\frac{x^2}{2x^2 + 1}$	$\frac{1}{2} + x^2$
------------------------	------------------------	------------------------	---------------------

(1 mark)

Q5**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6 3 marks

3 Simplify $\frac{x^2 - 16}{2x^2 - 5x - 12}$

(3 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6 6 marks

6 (a) Simplify fully $\frac{x^2 + 3x - 4}{2x^2 - 5x + 3}$

(3 marks)

(b) Write $\frac{4}{x+2} + \frac{3}{x-2}$ as a single fraction in its simplest form.

(3 marks)

Q7**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6

3 marks

11 Simplify fully $\frac{x^5 - 4x^3}{3x - 6}$

(3 marks)**Q8****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6

2 marks

10 Simplify $\frac{2^n - 1}{4^n - 1}$

(2 marks)

02 Multiply and Divide: KFC

Multiplying: **cancel first**, then multiply. Dividing: turn it into multiplying.

WHEN YOU SEE → YOU DO

“**x**” between fractions

→ cancel first

“**÷**” between fractions

→ Keep – Flip – Change



WORKED MODEL • KFC, THEN CANCEL

$$\frac{3x+6}{x^2-3x-10} \div \frac{x+5}{x^3-25x}$$

1 $\frac{3x+6}{x^2-3x-10} \times \frac{x^3-25x}{x+5}$

KFC

2 $\frac{3(x+2)}{(x-5)(x+2)} \times \frac{x(x+5)(x-5)}{x+5}$

factorise everything

3 cancel $(x+2)$, $(x-5)$, $(x+5)$

cancel first

ANSWER $3x$

IDEA 02

Practise: Multiply and Divide: KFC

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Cancel first

CH 2 · TOPIC 6

3 marks

Simplify $\frac{x^2 - 4}{3x + 6} \times \frac{9}{x - 2}$.

Answer _____

Q10**DR ESLAM**

KFC

CH 2 · TOPIC 6

3 marks

Simplify $\frac{2x + 6}{x^2 - 1} \div \frac{x + 3}{x - 1}$.

Keep · Flip · Change, then factorise and cancel.

Answer _____

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Easy

CH 2 · TOPIC 6 4 marks

1 (a) Simplify $3e^7f \times 4e^2f$

(2 marks)

(b) Simplify fully $(27a^{12})^{\frac{2}{3}}$

(2 marks)

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 2 marks

12 Simplify $\frac{25a}{8} \times \frac{2a}{5}$

Give your answer as a single fraction in its simplest form.

(2 marks)

Q13**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6 3 marks

- 2 Show that $\frac{1}{6x^2 + 7x - 5} \div \frac{1}{4x^2 - 1}$ simplifies to $\frac{ax + b}{cx + d}$ where a, b, c and d are integers.

(3 marks)

Q14**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6 4 marks

- 5 Show that $\frac{3x + 6}{x^2 - 3x - 10} \div \frac{x + 5}{x^3 - 25x}$ simplifies to ax where a is an integer.

(4 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2H • Q23

CH 2 • TOPIC 6 4 marks

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

03 Add and Subtract: the LCM

One fraction bar. The bottom is the LCM; each top is multiplied by what its bottom was missing.

WHEN YOU SEE → YOU DO

“as a single fraction”

→ LCM bottom, adjust the tops

“in its simplest form”

→ factorise the answer and cancel if you can

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

top × missing (x + 1)

top × missing (x - 2)

1

LCM of the bottoms

Every different bracket, letter and number — once each (after factorising).

2

Adjust the tops

Multiply each top by what its bottom was missing.

3

Expand and simplify the top

This is Expanding Brackets. Watch the minus in front of the second top.

WORKED MODEL

1
$$\frac{4(x + 1) - 3(x - 2)}{(x - 2)(x + 1)}$$

one fraction bar

2
$$4x + 4 - 3x + 6 = x + 10$$

expand the top

ANSWER
$$\frac{x + 10}{(x - 2)(x + 1)}$$

IDEA 03

Practise: Add and Subtract: the LCM

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Two brackets underneath

CH 2 · TOPIC 6 3 marks

Write $\frac{3}{x+2} + \frac{2}{x-1}$ as a single fraction.

Answer _____

Q17 DR ESLAM Numbers underneath · watch the minus

CH 2 · TOPIC 6 3 marks

Write $\frac{x+3}{4} - \frac{x-1}{6}$ as a single fraction.

Answer _____

SELECTED PRACTICE

Q18 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 3 marks

1 Write as a single fraction in its simplest form

$$\frac{2}{y+3} - \frac{1}{y-6}$$

(3 marks)

Q19 **PRACTICE** Practice · Medium

CH 2 · TOPIC 6 3 marks

8 Write $\frac{2x+1}{4} + \frac{x-2}{3}$ as a single fraction in its simplest form.

(3 marks)

Q20**PRACTICE**

Practice · Medium

CH 2 · TOPIC 6**2 marks**

16 Express as a single fraction.

$$\frac{m+1}{n+1} - \frac{m}{n}$$

Simplify your answer.

(2 marks)**Q21****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6**3 marks**

9 Express $\frac{1}{9x^2 - 25} - \frac{1}{6x + 10}$ as a single fraction in its simplest form.

(3 marks)

PAST PAPERS

Q22 **PAST PAPER** Edexcel 4MA1 · January 2022 · Paper 1HR · Q12**CH 2 · TOPIC 6** **5 marks****12** (a) Expand and simplify $n(n - 4)(3n + 5)$

(b) Express

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

as a single fraction in its simplest form.

(Total for Question 12 is 5 marks)**Q23** **PAST PAPER** Edexcel 4MA1 · November 2023 · Paper 1H · Q11**CH 2 · TOPIC 6** **3 marks****11** Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.**(Total for Question 11 is 3 marks)**

Q24 **PAST PAPER** Edexcel 4MA1 · May 2024 · Paper 1H · Q12**CH 2 · TOPIC 6** **5 marks**12 (a) Factorise fully $6y^2 - 5y - 4$ (b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

(Total for Question 12 is 5 marks)

Q25 **PAST PAPER** Edexcel 4MA1 · June 2025 · Paper 1H · Q12**CH 2 · TOPIC 6** **3 marks**12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

(Total for Question 12 is 3 marks)

04 Mixed Operations and Fractions Inside Fractions

BODMAS: brackets, then $\times$ and $\div$, and only then $+$ and $-$.

WHEN YOU SEE $\rightarrow$ YOU DO

" $4 - [\dots \div \dots]$ "

$\rightarrow$ BODMAS: bracket first, subtract last

"given $x = \dots$ and $y = \dots$, find x in terms of a "

$\rightarrow$ substitute, multiply top & bottom

1

Brackets and $\times \div$ first

Factorise, KFC, cancel — until one simple fraction is left.

2

Then $+$ $-$

Use the LCM with whatever is left. A whole term like 4 or $(x - 7)$ is a fraction over 1.

WORKED MODEL • JUN 2024 P2HR

$$4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$$

$$1 \quad (3x - 5) \times \frac{4x - 1}{(3x - 5)(x + 2)} = \frac{4x - 1}{x + 2}$$

KFC • factorise • cancel

$$2 \quad \frac{4(x + 2) - (4x - 1)}{x + 2}$$

now subtract

ANSWER $\frac{9}{x + 2}$

FRACTION INSIDE A FRACTION Substituting $y = \frac{5}{5a-2}$ into $\frac{5}{9y+5}$? Multiply the top and the bottom by the small denominator $(5a - 2)$ — the inner fraction disappears.

IDEA 04

Practise: Mixed Operations

My questions first, then selected practice, then real past papers.

Q26 **DR ESLAM** A whole number is a fraction over 1

CH 2 · TOPIC 6 2 marks

Write $2 - \frac{3}{x+1}$ as a single fraction.

Answer _____

Q27 **DR ESLAM** Fraction inside a fraction

CH 2 · TOPIC 6 3 marks

Given that $a = \frac{3}{x+1}$ and $x = \frac{2}{y}$, write a in terms of y as a single fraction.

Substitute, then multiply the top and bottom by y .

$a =$ _____

SELECTED PRACTICE

Q28**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6 4 marks

1 $2 - \frac{x+2}{x-3} - \frac{x-6}{x+3}$ can be written as a single fraction in the form $\frac{ax+b}{x^2-9}$

where a and b are integers.

Work out the value of a and the value of b .

(4 marks)

Q29**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 6 4 marks

4 Write

$$4 - \left[(x+3) \div \frac{x^2 + 5x + 6}{x-2} \right]$$

as a single fraction in its simplest form.
You must show your working.

(4 marks)

Q30**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 6 4 marks

6 Show that $6 + \left[(x+5) \div \frac{x^2 + 3x - 10}{x-1} \right]$ simplifies to $\frac{ax-b}{cx-d}$ where a , b , c and d are integers.

(4 marks)

Q31**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 6 4 marks

16 Write $\frac{5}{4x^2 - 25} - (2x + 3) \div \left(\frac{4x^2 + 16x + 15}{7} \right)$ as a single fraction in its simplest form.

Show clear algebraic working

(4 marks)**PAST PAPERS****Q32****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q24

CH 2 · TOPIC 6 4 marks**24** Express

$$\left(\frac{4}{2x - 5} - \frac{3}{2x - 3} \right) \div \frac{9x - 4x^3}{6x^2 - 17x + 5}$$

as a single fraction in its simplest form.

(Total for Question 24 is 4 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q21

CH 2 • TOPIC 6 5 marks

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

.....

(Total for Question 21 is 5 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q21

CH 2 • TOPIC 6 4 marks

21 Write $\frac{25x^2-64}{5x^2-13x-6} \times \frac{x^2-8x+15}{5x+8} - (x-7)$

as a single fraction in its simplest form.
Show clear algebraic working.

.....

(Total for Question 21 is 4 marks)

Q35 **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 1H • Q21**CH 2 • TOPIC 6** **4 marks**

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

Q36 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 2HR • Q23**CH 2 • TOPIC 6** **3 marks**

23 Express $\left(\frac{20}{x^2-36} - \frac{2}{x-6}\right) \times \frac{1}{4-x}$ as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

Q37 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 2H · Q19

CH 2 · TOPIC 6 3 marks

$$19 \quad a = \frac{14}{3x-7} \quad x = \frac{7}{4y-3}$$

Express a in the form $\frac{py+q}{ry+s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a = \dots\dots\dots$

(Total for Question 19 is 3 marks)

Q38 PAST PAPER Edexcel 4MA1 · May 2023 · Paper 1HR · Q23

CH 2 · TOPIC 6 4 marks

$$23 \quad \text{Simplify } (x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x} \right) - 2x$$

Give your answer in the form $\frac{ax^2}{bx+c}$ where a, b and c are integers.

$\dots\dots\dots$

(Total for Question 23 is 4 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q26

CH 2 · TOPIC 6 4 marks

26 Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2H · Q20

CH 2 · TOPIC 6 3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x + 2}$ in the form $\frac{a - by^2}{c + dy}$ where a, b, c and d are integers.

(Total for Question 20 is 3 marks)

Q41 PAST PAPER Edexcel 4MA1 · November 2024 · Paper 2H · Q23

CH 2 · TOPIC 6 4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

Q42 PAST PAPER Edexcel 4MA1 · November 2025 · Paper 1H · Q23

CH 2 · TOPIC 6 3 marks

23 $a = \frac{2x + 5}{1 - x}$ $x = \frac{5 - 2y}{3y}$

Write a in the form $\frac{m + ny}{p(y - 1)}$ where m , n and p are integers.

Show your working clearly.

$$a = \dots\dots\dots$$

(Total for Question 23 is 3 marks)

05 Solving Equations with Fractions

An equation lets you get rid of every denominator at once: multiply **every term** by the LCM.

WHEN YOU SEE → YOU DO

“solve” + fractions

→ × every term by the LCM

“give your answers to 3 s.f.”

→ ERCS → quadratic formula

a speed or a length

→ reject the negative answer

1

Clear the fractions

Multiply every term (the right-hand side too) by the LCM. Two fractions equal? Cross-multiply.

2

ERCS

Expand, Rearrange to = 0, Collect → $ax^2 + bx + c = 0$ (or a linear equation).

3

Solve and check

Calculator first. Reject any answer that makes a denominator zero — or a speed negative.

WORKED MODEL • JAN 2021 P2HR

$$\frac{5}{x+2} + \frac{3}{x(x+2)} = 2$$

1 $5x + 3 = 2x(x + 2)$

× LCM $x(x + 2)$

2 $2x^2 - x - 3 = 0$

ERCS

ANSWER $x = \frac{3}{2}$ or $x = -1$

CALCULATOR CHECK Put $x = 1.5$ back in: $\frac{5}{3.5} + \frac{3}{5.25} = 2$. Correct.

IDEA 05

Practise: Solving Equations with Fractions

My questions first, then selected practice, then real past papers.

Q43 DR ESLAM × LCM, linear

CH 2 · TOPIC 6 3 marks

Solve $\frac{x+1}{3} - \frac{x-2}{4} = 2$.

x = _____

Q44 DR ESLAM × LCM, then a quadratic

CH 2 · TOPIC 6 4 marks

Solve $\frac{3}{x-1} + \frac{2}{x+1} = 1$, giving your answers to 2 d.p.

× $(x-1)(x+1)$, then ERCS, then the calculator.

x = _____

SELECTED PRACTICE

Q45**PRACTICE**

Practice · Medium

CH 2 · TOPIC 6 4 marks

17 Solve $\frac{3x-2}{5} - \frac{3-4x}{2} = 2$

Show clear algebraic working.

(4 marks)

Q46**PRACTICE**

Practice · Hard

CH 2 · TOPIC 6 3 marks

15 Solve

$$\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$$

Show clear algebraic working

(3 marks)

Q47 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 6 4 marks

13 Solve $\frac{x}{x+4} + \frac{7}{x-2} = 1$

You **must** show your working.

$X = \dots\dots\dots$

(4 marks)

Q48 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 6 6 marks

15 Solve this equation, giving your answers correct to 1 decimal place.

$$\frac{5}{x+2} + \frac{3}{x-3} = 2$$

$X = \dots\dots\dots$ or $X = \dots\dots\dots$

(6 marks)

PAST PAPERS

Q49**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q18

CH 2 • TOPIC 6 5 marks**18** Solve the equation

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

Show clear algebraic working.

.....

(Total for Question 18 is 5 marks)

Q50**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q24

CH 2 • TOPIC 6 5 marks

24 Solve $\frac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

(Total for Question 24 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--|--|
| 00 | Golden rule | Factorise every top and every bottom first. |
| 01 | Simplify | Cancel whole brackets only. $(3 - x) = -(x - 3)$. |
| 02 | $\times$ and $\div$ | Cancel first. Dividing: Keep – Flip – Change. |
| 03 | $+$ and $-$ | LCM bottom $\cdot$ adjust the tops $\cdot$ expand the top. |
| 04 | Mixed | BODMAS: $\times \div$ first, $+-$ last. Nested: multiply top & bottom. |
| 05 | Equations | $\times$ LCM every term $\rightarrow$ ERCS $\rightarrow$ solve $\rightarrow$ reject bad answers. |

BOTTOM LINE Factorise $\rightarrow$ cancel $\rightarrow$ KFC for $\div$ $\rightarrow$ LCM for $\pm$ $\rightarrow$ in an equation, multiply every term by the LCM.

Answers

Algebraic Fractions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SIMPLIFYING: CANCEL BRACKETS

Q1	Dr Eslam • Golden rule first	$\frac{x-3}{x-2}$
Q2	Dr Eslam • HCF on top	$\frac{x}{x-5}$
Q3	Practice • Medium	$\frac{4}{x-3}$
Q4	Practice • Medium	$\frac{x^2}{2x^2+1}$
Q5	Practice • Hard	$\frac{x+4}{2x+3}$
Q6	Practice • Hard	(a) $\frac{x+4}{2x-3}$ (b) $\frac{7x-2}{(x+2)(x-2)}$
Q7	Practice • Hard	$\frac{x^3(x+2)}{3}$
Q8	Practice • Very Hard	$\frac{1}{2^n+1}$

IDEA 02 • MULTIPLY AND DIVIDE: KFC

- Q9** Dr Eslam • Cancel first 3
- Q10** Dr Eslam • KFC $\frac{2}{x+1}$
- Q11** Practice • Easy (a) $12e^9f^2$ (b) $9a^8$
- Q12** Practice • Medium $\frac{5a^2}{4}$
- Q13** Practice • Very Hard $\frac{2x+1}{3x+5}$
- Q14** Practice • Very Hard $3x$
- Q15** Edexcel 4MA1 • June 2025 • Paper 2H • Q23 $p = 8$

IDEA 03 • ADD AND SUBTRACT: THE LCM

Q16	Dr Eslam • Two brackets underneath	$\frac{5x + 1}{(x - 1)(x + 2)}$
Q17	Dr Eslam • Numbers underneath • watch the minus	$\frac{x + 11}{12}$
Q18	Practice • Medium	$\frac{y - 15}{(y + 3)(y - 6)}$
Q19	Practice • Medium	$\frac{10x - 5}{12}$
Q20	Practice • Medium	$\frac{n - m}{n(n + 1)}$
Q21	Practice • Very Hard	$\frac{7 - 3x}{2(9x^2 - 25)}$
Q22	Edexcel 4MA1 • January 2022 • Paper 1HR • Q12	(a) $3n^3 - 7n^2 - 20n$ (b) $\frac{3x + 16}{4x}$
Q23	Edexcel 4MA1 • November 2023 • Paper 1H • Q11	$\frac{7x + 10}{12x}$
Q24	Edexcel 4MA1 • May 2024 • Paper 1H • Q12	(a) $(2y + 1)(3y - 4)$ (b) $\frac{31 - 14x}{12x}$
Q25	Edexcel 4MA1 • June 2025 • Paper 1H • Q12	$\frac{17x - 6}{12x}$

IDEA 04 • MIXED OPERATIONS

- Q26** Dr Eslam • A whole number is a fraction over 1 $\frac{2x-1}{x+1}$
- Q27** Dr Eslam • Fraction inside a fraction $\frac{3y}{y+2}$
- Q28** Practice • Very Hard $a = 4, b = -42$
- Q29** Practice • Very Hard $\frac{3x+10}{x+2}$
- Q30** Practice • Very Hard $\frac{7x-13}{x-2}$
- Q31** Practice • Very Hard $\frac{40-14x}{(2x+5)(2x-5)}$
- Q32** Edexcel 4MA1 • January 2020 • Paper 2H • Q24 $\frac{1-3x}{x(2x-3)^2}$
- Q33** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q21 $\frac{34-6x}{(x-5)(x-1)}$
- Q34** Edexcel 4MA1 • January 2021 • Paper 2H • Q21 $\frac{54}{5x+2}$
- Q35** Edexcel 4MA1 • May 2021 • Paper 1H • Q21 $x = \frac{5a-2}{5a+7}$
- Q36** Edexcel 4MA1 • June 2022 • Paper 2HR • Q23 $\frac{2}{(x+6)(x-6)}$
- Q37** Edexcel 4MA1 • June 2022 • Paper 2H • Q19 $a = \frac{3-4y}{2y-3}$
- Q38** Edexcel 4MA1 • May 2023 • Paper 1HR • Q23 $\frac{-7x^2}{4x+1}$
- Q39** Edexcel 4MA1 • June 2024 • Paper 2HR • Q26 $\frac{9}{x+2}$
- Q40** Edexcel 4MA1 • June 2024 • Paper 2H • Q26 $\frac{5-20y^2}{1+8y}$

$$4(2x-3) - (7+8x) = -19, n = -19$$

IDEA 05 · SOLVING EQUATIONS WITH FRACTIONS

- Q43** Dr Eslam · × LCM, linear $x = 14$
- Q44** Dr Eslam · × LCM, then a quadratic $x = \frac{5 \pm \sqrt{33}}{2}$: 5.37 or -0.37
- Q45** Practice · Medium $x = \frac{3}{2}$
- Q46** Practice · Hard $y = 2.9$
- Q47** Practice · Very Hard $x = -12$
- Q48** Practice · Very Hard $x = 5.3$ or -0.3
- Q49** Edexcel 4MA1 · January 2021 · Paper 2HR · Q18 $x = \frac{3}{2}$ or $x = -1$
- Q50** Edexcel 4MA1 · November 2023 · Paper 2H · Q24 $x = \frac{7}{4}$